A CHARACTERISATION OF BOUNDED PROJECTIVE AMPLITUDE FOR A COMPLEX

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Projective amplitude of a complex and the notion of a bounded representative. defined in the same section, source lines 17360-17412, before the counted statement. Supplies both sides of the characterisation.
- Cohomology of a complex can be killed degree by degree by choosing a projective module mapping onto it. used at proof lines 17430-17460. Constructs the resolution one degree at a time.
- Termination of the construction in the bounded case. used at proof lines 17460-17480. Shows that the resolution is bounded, which is what the characterisation requires.

Source reference. Original lemma 0A5P: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 12817–12817. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 05T7: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7471–7471. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 05CF: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6918–6918. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.2. *Let R be a ring. Let K be an object of $D(R)$. Let $a, b \in \mathbf{Z}$. The following are equivalent*

1. *K has projective-amplitude in $[a, b]$,*
2. *$\mathrm{Ext}^i_R(K, N) = 0$ for all R-modules N and all $i \notin [-b, -a]$,*
3. *$H^n(K) = 0$ for $n > b$ and $\mathrm{Ext}^i_R(K, N) = 0$ for all R-modules N and all $i > -a$, and*
4. *$H^n(K) = 0$ for $n \notin [a-1, b]$ and $\mathrm{Ext}^{-a+1}_R(K, N) = 0$ for all R-modules N.*

Proof. Assume (1). We may assume K is the complex

$$\ldots \to 0 \to P^a \to P^{a+1} \to \ldots \to P^{b-1} \to P^b \to 0 \to \ldots$$

where P^i is a projective R-module for all $i \in \mathbf{Z}$. In this case we can compute the ext groups by the complex

$$\ldots \to 0 \to \mathrm{Hom}_R(P^b, N) \to \ldots \to \mathrm{Hom}_R(P^a, N) \to 0 \to \ldots$$

and we obtain (2).

Assume (2) holds. Choose an injection $H^n(K) \to I$ where I is an injective R-module. Since $\mathrm{Hom}_R(-, I)$ is an exact functor, we see that $\mathrm{Ext}^{-n}(K, I) = \mathrm{Hom}_R(H^n(K), I)$. We conclude in particular that $H^n(K)$ is zero for $n > b$. Thus (2) implies (3).

By the same argument as in (2) implies (3) gives that (3) implies (4).

Assume (4). The same argument as in (2) implies (3) shows that $H^{a-1}(K) = 0$, i.e., we have $H^i(K) = 0$ unless $i \in [a, b]$. In particular, K is bounded above and we can choose a a complex $P^\bullet$ representing K with P^i projective (for example free) for all $i \in \mathbf{Z}$ and $P^i = 0$ for $i > b$. See Derived Categories, Lemma 05T7. Let $Q = \mathrm{Coker}(P^{a-1} \to P^a)$. Then K is quasi-isomorphic to the complex

$$\ldots \to 0 \to Q \to P^{a+1} \to \ldots \to P^b \to 0 \to \ldots$$

as $H^i(K) = 0$ for $i < a$. Denote $K' = (P^{a+1} \to \ldots \to P^b)$ the corresponding object of $D(R)$. We obtain a distinguished triangle

$$K' \to K \to Q[-a] \to K'[1]$$

in $D(R)$. Thus for every R-module N an exact sequence

$$\mathrm{Ext}^{-a}(K', N) \to \mathrm{Ext}^1(Q, N) \to \mathrm{Ext}^{1-a}(K, N)$$

By assumption the term on the right vanishes. By the implication (1) $\Rightarrow$ (2) the term on the left vanishes. Thus Q is a projective R-module by Algebra, Lemma 05CF. Hence (1) holds and the proof is complete. $\square$